RETAILOS / RETROSPECTIVE DECISION STUDY

Evidence before complexity.

Can better sales forecasts improve inventory decisions under a shared purchasing budget?

42
product–store series

17
28-day evaluation episodes

4
ordering policies

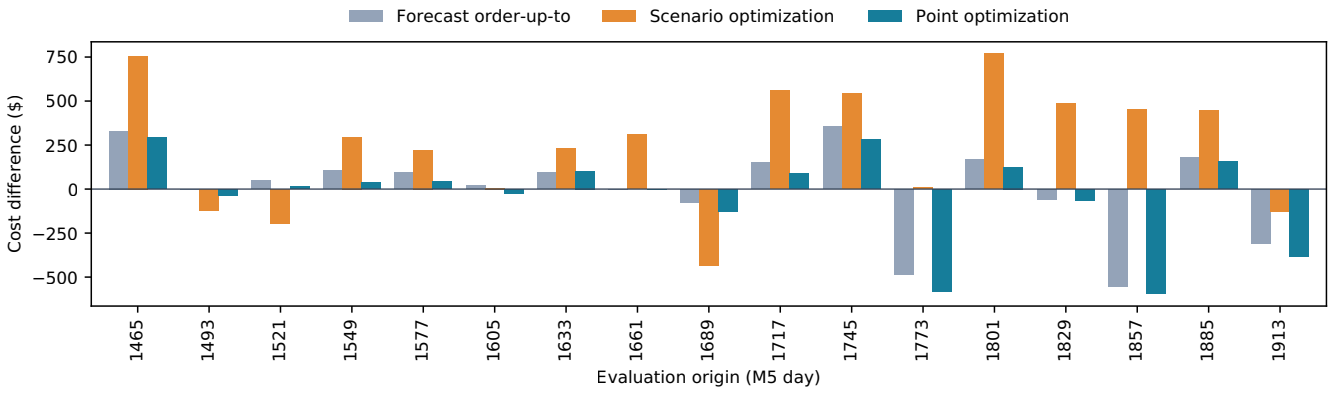
> **Decision:** retain trailing-average ordering as the practical benchmark. Point optimization has a small aggregate cost advantage, but wins only 8 of 17 windows and has a lower unit fill rate. Scenario optimization has not established added value.

1. What the comparison found

Policy	Modeled cost	Cost change*	Lower-cost windows*	Proxy fill rate
Trailing-average orders	$131,805.88	+0.00%	—	79.87%
Forecast order-up-to	$131,858.29	+0.04%	7/17	79.06%
Point optimization	$131,127.83	-0.51%	8/17	79.35%
Scenario optimization	$136,010.21	+3.19%	4/17	77.60%

*Against trailing-average orders. Negative change means lower cost. All costs are simulations under assumed operating inputs; no actual Walmart savings were measured.

2. Small aggregate gains can hide uneven performance



Each policy has a separate bar; differences are not added together. Below zero means lower cost than trailing-average orders. Period: February 2, 2015–May 22, 2016.

What I compared. What I would do next.

3. Keep the comparison fair

I compared trailing-average and forecast order-up-to rules with point and scenario optimization. Both optimizers use the same objective and constraints; the point optimizer uses one forecast path and the scenario optimizer uses forecast-plus-residual paths. Within each episode, all policies share initial stock, lead time, purchasing budget and evaluation sales.

4. Read cost alongside consistency and service

Point optimization cost $678.05 less than trailing-average orders in aggregate, but won only 8 of 17 periods and had a lower unit fill rate (79.35% versus 79.87%). It beat scenario optimization in 14 of 17 periods. I would retain the simple benchmark while investigating the scenario construction, rather than conclude that uncertainty-aware optimization is generally inferior.

5. Distinguish prediction from uncertainty quality

Gradient boosting had daily MAE of 1.645 versus 1.712 for the trailing mean, about 3.95% lower. Twelve earlier residual blocks supported eight sampled paths. The daily 10-90% scenario envelope covered 52.42% of evaluation observations. I treat this as an empirical range, not a calibrated 80% prediction interval.

6. Check what the verification establishes

A separate daily replay of saved orders matched all 68 policy-period costs, with maximum absolute difference below $0.000001. It checked integer orders, budget feasibility, unmet units and ending stock. This supports output-accounting consistency; it does not repeat training from the official raw files.

7. Define the next evaluation

This is a retrospective exploratory comparison, including previously inspected periods. Episode inventory resets and shared histories limit interpretation. Only 3 of 42 series overlap the initial study, so I keep the studies separate. Observed sales may understate demand; costs, penalties and initial stock are assumed, with no terminal salvage credit. I have not measured actual retailer savings or established statistical significance.

What I want to study next

I want to understand how forecasting, optimization and service objectives should jointly guide a decision under limited resources. My next evaluation would use verified operating inputs, a policy fixed in advance and prospective outcomes before recommending adoption.

Source: official M5 data from my local study. Aggregate evidence and development notes:
github.com/younjuhan03/retail-decision-intelligence